Observed source oddities and improvement opportunities

Prepared 2026-10-29 for the project owner. Tracking: #251 and #255.

The highest-impact producer-side opportunity is the catalog generator: 992 automated Rules across 21 packages have complete embedded check content, but their new catalog entries use the wrong namespace for standard URI bindings. Several smaller, repeatable authoring problems are also suitable for generator checks. The integer-to-concat failures need a standards decision before they should be called content bugs. SV-215833 deserves a focused authoring review.

This report covers all oddity classes identified by the existing pinned 65-benchmark conversion and follow-up investigations. It is not a new exhaustive semantic audit of every original check, and it makes no scanner-runtime claims. No source content, validator, schema, compiler or scanner was changed to produce it. Suggested areas of control are hypotheses for you to confirm, not assumptions about which artifacts you own.

How to use this report

The companion **affected-rules.csv** is the working list. Filter `finding_ids` for an O-number below, then use the exact source filename, Rule ID, original diagnostic and pinned source link. It includes 3,024 distinct Rules: all 2,981 currently blocked automated Rules, five additional manual Rules with the Switch ID anomaly, and 38 IOS XE Rules whose tooling limitation is now resolved.

findings.json retains the detailed classification and 71 original closure validation records. The PDF is a printable copy of this report. All three are included in the accompanying report package; the source originals are also preserved in the complete DRAFT archive.

Finding counts overlap. For example, one Rule can have a wrong reference type, a deprecated Object, and a bad pattern operation. Counts are affected Rules, not unique source nodes or guaranteed conversion gains.

Where you may be able to improve the producing pipeline

Priority	Finding	Observed reach	Likely place to investigate
First	O01: foreign namespace on catalog URI entries	992 automated Rules, 21 packages	Catalog/XML element creation
First	O02: stale standard catalog references	20 dangling entries in 20 packages; 979 of those 992 Rules share these packages	Enhancement/repackaging graph construction
First	O03: missing record datatype	11 investigated Rules	Cmdlet Object generator
First	O04: pattern paths with prohibited recursion behavior	7 investigated Rules	File acquisition templates
First	O05: missing/incompatible operation on pattern entities	11 investigated Rules	Regex acquisition templates
First	O06: Test-level absence with a State reference	4 investigated Rules	Test construction/requirement modeling
First	O07: wrong Object/State/Filter family	4 investigated IIS Site Rules	Test family and reference generation
Review	O11/O12: IOS XE text-as-int and delimiter	SV-215833	Timeout-check Variable pipeline
Review	O08: deprecated constructs	162 observed Rules, including 151 primary effective-Test exclusions	Supported-content maintenance
Review	O09: nonstandard sqlext packages	137 automated Rules in four SQL Server packages	Extension publishing/compatibility
Cleanup	O13: Switch IDs carry ms.dns prefix	All 42 Rules in the IOS XE Switch NDM package	Identifier/provenance template

The small validation counts above are observed counts within investigated closures, not a claim that the other content is free of the same pattern.

O01/O02 — catalog entries and stale references

Every one of the 992 requested Definitions is present in exactly one embedded OVAL component. The intended source chain is also present: URI entry, matching component-ref, component, exact Definition ID. The problem is the entry's QName:

```
<!-- Observed extension element: SCAP namespace -->
<ds:uri name="...-oval2.xml" uri="#..._cref_...-oval2.xml"/>

<!-- Standard XML Catalog entry, shown as an improvement target -->
<cat:uri name="...-oval2.xml" uri="#..._cref_...-oval2.xml"/>
```

`ds` denotes `http://scap.nist.gov/schema/scap/source/1.2`. `cat` denotes `urn:oasis:names:tc:entity:xmlns:xml:catalog`. This is a namespace-URI issue, not a choice of prefix spelling. Changing a prefix while keeping the SCAP namespace would not fix the standard binding.

The catalog XSD allows foreign elements with `namespace="##other"` and `processContents="skip"`. These entries can therefore pass structural XSD validation while contributing no standard XML Catalog URI mapping. This is why an XSD-only packaging check can miss the problem.

Twenty affected packages also retain one older standard catalog entry whose component-ref points to a missing old component. The remaining affected package, Windows 2012/2012 R2 DC, has a usable old catalog but lacks standard bindings for its new foreign entries. The 979 figure is affected Rules sharing the 20 stale-entry packages; it does not mean 979 distinct dangling entries.

Improvement: create new entries using the XML Catalog namespace, then check every standard URI binding, component-ref target, selected check href and Definition ID after repackaging. Reconcile stale entries with the intended final component graph. Keep publisher signatures and provenance coherent when issuing a corrected package. Do not silently rewrite previously signed/frozen archives.

No missing requested Definition or relative-path normalization defect was found in this 992-Rule cohort. Correct bindings may expose additional content or converter blockers; 992 is not a promised unlock count.

O03 — nil cmdlet parameters still need the record datatype

Eleven investigated Windows Rules fail the original assertion that a `cmdlet_object` parameters entity must have datatype `record`. The relevant templates emit a nil parameters entity without the required record type.

```
<!-- Observed shape -->
<parameters xsi:nil="true"/>

<!-- Type annotation to review; intended nil semantics must stay intact -->
<parameters datatype="record" xsi:nil="true"/>
```

Nil does not waive the datatype constraint. The affected sources span Windows 10, Windows 11, and Server 2016/2019/2022/2025. Review the common generator rather than patching eleven unrelated checks by hand. The exact Objects and Rules are listed under O03 in the CSV.

O04 — path patterns and recursion behaviors conflict

Seven investigated Rules combine a pattern-match path entity with `max_depth` and/or `recurse_direction`, which the pinned original Schematron prohibits. Examples include .NET SV-225229 and Windows 11 SV-253280, with analogous checks in Windows 10 and Server 2016/2019/2022/2025.

Improvement: determine whether the intended search is represented by the path regex or by traversal from a selected directory, then author a permitted form. Simply dropping recursion could change the collection population. A useful generator check is to reject this combination before the signed package is built.

O05 — pattern entities have incompatible or omitted operations

Eleven investigated Rules fail assertions requiring a pattern entity's operation to be `pattern match`. They occur in Firefox Windows, .NET, Oracle Linux 7/8/9 and RHEL 8/9. Some original assertions report the same node under both shellcommand/textfilecontent54 descriptions; the count is distinct Rules, not the number of assertion messages.

Improvement: verify that the entity really contains a regex and explicitly emit `operation="pattern match"` where the capability requires it. Keep this separate from comparisons on returned State fields. Do not globally replace `equals` operations on arbitrary string entities.

O06 — Test-level none_exist also references a State

Four investigated Rules violate the original rule that a Test with `check_existence="none_exist"` must not reference a State:

- Oracle Linux 9 SV-271451r1137691.
- RHEL 9 SV-257789r1137691.
- IIS 8.5 Server SV-258446r928857.
- IIS 8.5 Site SV-258445r928856.

Improvement: resolve the intended condition. "No selected Items exist" and "selected Items exist but none match a State" are different requirements. Removing the State or changing existence without that review can change meaning. Field-level absence is also a different scope; see O20 below.

O07 — Object, State and Filter types do not agree

IIS 8.5 Site SV-214451r879567, SV-214452r879568, SV-214463r879643 and SV-214495r879587 contain original family/reference validation failures. The diagnostics include `appcmd_test` references that do not resolve to the required `appcmd_object`/`appcmd_state` family and Filter States of the wrong type. Some of these Rules also contain deprecated content.

Improvement: validate the typed reference graph, not just ID existence. Every referenced node must have the kind and capability family required by its consumer. A rename of the XML element alone is not a proven correction: different capabilities can return different fields and have different collection behavior.

O08/O09 — deprecated constructs and nonstandard SQL extensions

The primary ledger excludes 151 Rules for effectively deprecated Tests. Detailed validation of the 71 investigated closures finds 11 additional Rules with deprecated Objects, States, Tests or attribute values, giving **162 distinct observed Rules** in O08. This is an overlap-aware inventory of different kinds of deprecation; it is not 162 deprecated Test types. Examples include legacy Windows user/access-token/WMI constructs and deprecated Test check values.

Improvement: keep a versioned support inventory, including explicit later OVAL reinstatement decisions, and migrate deprecated constructs only to reviewed semantic replacements. Do not treat every historical `deprecated_info` annotation as current deprecation if a recorded reinstatement applies. Do not turn old `check="none exist"` into an arbitrary modern quantifier without checking intent.

Four SQL Server packages use nonstandard `independent.sqlext` content: SQL Server 2016 Database/Instance and SQL Server 2022 Database/Instance. Their 137 automated Rules are held at the package-extension boundary. This does not establish that their checks are logically incorrect.

Improvement: if maintaining that extension, publish its versioned grammar, collector/result semantics and compatibility requirements alongside the content. Using a standard SQL capability is an option only where equivalence is reviewed; do not assume that changing the element name preserves behavior.

IOS XE findings: authoring concerns versus standards conflicts

The four Router NDM validation failures below occur in the pinned NIWC enhanced `oval2` component with `oval:navy.navwar.niwcatlantic.scc.iosxe:*` identities. The active XCCDF checks point to that component. Attribution to DISA's original content has not been established.

O10 — integer-to-concat is a disputed validation boundary

SV-215807r1186318, SV-215812r1137875, SV-215833r961068 and SV-215854r1156415 insert an integer count into string construction. Their apparent intent is to format a count inside a regex repetition quantifier. The pinned Schematron rejects Variable references inside concat unless the referenced Variable is declared string.

The same schema's FunctionGroup prose says a cast should be attempted before reporting an error, and explicitly illustrates an integer cast to string for concat. Its example names a registry component, whereas the failing assertions restrict Variable references. That exact scope needs standards review.

Improvement opportunity: reconcile the casting prose and validator, and test the intended count-to-regex pipeline. These are confirmed assertion failures; three of the Rules should not be labeled confirmed content bugs on that basis alone. Do not change count datatypes or remove assertions just to obtain a pass.

O11/O12 — SV-215833 has stronger authoring concerns

Variable `...:var:21583304` is declared `int` but described as splitting the configuration into lines. Variable `...:var:21583307` is also declared `int` and uses `regex_capture pattern="\s+exec-(.*)"` on that text. A representative capture from `exec-timeout 5 0` would be `timeout 5 0`, not an integer. That example is constructed from the expression; no live scanner result is claimed.

The split delimiter is the literal two-character XML string `\n` (code points 92 and 110), not a newline character. OVAL documents a delimiter string. Reliance on scanner-specific escape interpretation is unverified.

Numeric captures are valid. This same Rule has a separate HTTP-timeout Variable `...:var:21583310` declared `int`, using `regex_capture pattern="ip http timeout-policy idle (\d+)"`. A capture such as 5 is a valid integer value for numeric comparison with 300. That branch is not the text-datatype concern. The concern is confined to intermediate Variables 21583304 (split configuration text), 21583307 (capture `timeout 5 0`) and 21583305 (merge timeout lines), each declared `int`. The `(5|4|3|2|1)` group in the later State regex checks the merged text; it does not change what the preceding `regex_capture` Variable returns.

Improvement: review those text-bearing intermediate declarations, all downstream consumers, and the delimiter's actual lexical value. Keep numeric capture Variables numeric where their captured values are integer-representable. Add examples with multiple configuration lines and nonmatching/empty output. This is a stronger content-defect candidate than formatting an integer count as string, but no source correction or runtime equivalence is established.

O13 — Switch IDs look copied from DNS content

All 42 Rules in IOS XE Switch NDM enhancedV6 use an `ms.dns` Rule-ID prefix; 37 are automated and five are manual/OCIL-only. This is a confirmed provenance or template anomaly, not proof that the associated checks evaluate DNS logic.

Improvement: review the identifier template for future output. Treat existing IDs as published identities: changing them requires updating references and maintaining migration/provenance records rather than casual renaming.

Router SV-215838r1015295's NTP-server `count` is a separate converter limitation (O17). The 37 Switch automated Rules are in the catalog group O01. The remaining 38 Router Rules have now passed independent static verification (O23); their earlier failure was our missing reader crosswalk, not a demonstrated source bug.

Valid source constructs that should not be changed to appease conversion

Finding	Primary affected Rules	What is actually unresolved
O14: Test all_exist	1,184	Missing expected populations, error/unknown/incomplete evidence and native existence semantics
O15: check imports/exports	381	Source-bound external/organizational input and binding recovery
O16: multiple aggregation components	30	Collection-expression/cardinality/dependency semantics
O17: count/split/substring	33	Function support: 15 count, 15 split, three substring
O18: acquisition comparison/datatype	2	Case-insensitive command and version-typed module acquisition
O19: non-integer ordered comparison	10	Converter's integer-only expansion boundary; original closures validated
O20: comparator-bearing field absence	9	Absence-versus-comparison normalization requires source-intent review
O21: source fields missing from crosswalk	2	rpmverifyfile ghost_file/configuration_file field representation
O22: Variable-source attributes/function chain	2	Valid Solaris source attributes and subsequent split support

These are conversion/contract boundaries, not an instruction to simplify valid OVAL. Secondary blockers overlap, and most blocked content has not received independent semantic verification. The listed construct can be supported by OVAL while still needing SCAP-NG implementation or contract work.

For O14, a complete successful population of actual Items makes all_exist and one_or_more agree, but a positive Item does not establish complete accounting of all expected targets. The real Amazon Linux 2023 SV-274101 example selects seven audit-tool paths through a Variable. Preserve missing-target, error, unknown and incomplete distinctions; do not globally replace the source token.

For O18, the two validated originals are .NET SV-225226r961041 (case-insensitive command acquisition) and Windows 10 SV-220812 (version-typed module acquisition). Do not erase the operation or cast version to string merely to fit a target.

For O20, examples require an absent stdout/stderr field but also carry a literal `.*` comparator. That may be redundant authoring, but field absence is not the same as "a field exists and no values match". The prior

owner-approved exact normalization remains bounded; it is not permission for a global rewrite.

For O21, the affected valid originals are RHEL 7 SV-214799r991589 and Oracle Linux 7 SV-221653r1015161. Dropping ghost/configuration-file filters could change which files are assessed. This needs mapping review, not source deletion.

Practical checks for future published content

If these are within your control, the most useful additions to the producing pipeline would be:

1. **A semantic reference-closure check after enhancement and packaging.** Resolve every selected check href through standard catalog entries to the expected Definition, and verify component-ref targets and typed Object/State/Filter references. Flag foreign catalog-like elements separately from real mappings.

2. **Original XSD and Schematron checks with per-Rule attribution.** Record exact diagnostics; distinguish invalid content, deprecated content and known standards/validator conflicts. A green XSD-only run is insufficient.

3. **Template checks for the recurring patterns above.** Record datatypes on nil cmdlet parameters, path-pattern/recursion combinations, required pattern operations, forbidden none_exist/State combinations, and meaningful ID prefixes.

4. **Small Variable/function examples with expected values and failures.** Include count-to-string formatting, text captures, literal delimiters, empty output, missing inputs, unknown and errors. Keep observed scanner results distinct from static validation or illustrative examples.

5. **A support and extension inventory.** Identify effective deprecations and reinstatements, and provide the normative contract for vendor extensions.

6. **Versioned corrections.** Reissue corrected source with new identity/version and checksums as appropriate, preserving earlier signed artifacts and a concise explanation of what changed and whether evaluation meaning changed.

This report recommends checks; it does not install a new CI gate or change the existing source-validation policy.

Evidence and accounting

Pinned production inputs: `niwc-atlantic/scap-content-library` revision `8c8e5dff860af6b1290ee9273a282db24278f8d5`. Implementation/recovery record: `vanderpol/scap-ng` at `2e5eaec0caf031d792ea6dd996a649e48a91c24d`. The report reuses existing ledgers and validation records; it does not rerun the 65-source conversion.

• 65 benchmarks, 8,892 original Rules, 7,239 automated attempts.

• Separate published static proof total: 4,258 primary automated Rules.

• Residual automated blockers: 2,981. Manual/OCIL-only Rules preserved: 1,653.

• The original complete DRAFT still labels 52 candidates UNVERIFIED; their later

static proofs are published separately. Neither label implies live runtime parity.

• The 71 detailed closure records combine 32 recorded exclusions and 39 additional

investigations. They are not a validity census of all 7,239 automated Rules.

Durable supporting records:

• Complete original-preserving DRAFT.

• Separate 52-proof checkpoint.

• Full population/catalog/IOS XE investigation.

• IOS XE exact original excerpts.

• Population/casting/catalog decision packet.

The accompanying CSV and JSON provide the affected source filenames, Rule IDs, hashes, confidence and suggested control area for every indexed Rule. Corrections to NIWC-enhanced artifacts should not be attributed to DISA originals without a separate authoritative-source comparison.

Appendix A — all 21 catalog-affected packages

Pinned package	Automated Rules	Stale standard entry
Active Directory Domain V3R6 / enhancedV2	17	Yes
Active Directory Forest V3R2 / enhancedV2	4	Yes
Apache Server 2-4 UNIX Server V3R2 / enhancedV2	22	Yes
Apache Server 2-4 UNIX Site V2R6 / enhancedV2	6	Yes
Apache Server 2-4 Windows Server V3R3 / enhancedV2	23	Yes
Apache Server 2-4 Windows Site V2R2 / enhancedV2	5	Yes
Apple macOS 13 V1R5 / enhancedV7	102	Yes
Apple macOS 14 V2R3 / enhancedV7	152	Yes
Apple macOS 15 V1R4 / enhancedV8	157	Yes
Apple macOS 26 V1R1 / enhancedV5	157	Yes
Cisco IOS-XE Switch NDM V3R5 / enhancedV6	37	Yes
Cisco IOS Router NDM V3R6 / enhancedV6	29	Yes
Cisco IOS Switch NDM V3R6 / enhancedV6	29	Yes
F5 NGINX V1R1 / enhancedV1	22	Yes
MS Exchange 2019 Edge Server V2R2 / enhancedV1	55	Yes
MS Exchange 2019 Mailbox Server V2R3 / enhancedV1	51	Yes
MS IIS 10-0 Server V3R6 / enhancedV18	25	Yes
MS IIS 10-0 Site V2R14 / enhancedV19	33	Yes
MS OneDrive V2R4 / enhancedV12	12	Yes
MS Windows 2012 and 2012 R2 DC V3R6 / enhancedV13	13	No
MS Windows Server DNS V2R3 / enhancedV8	41	Yes

Appendix B — smaller source-validation groups to review

Full versioned Rule IDs are shown below; the CSV retains the complete XML IDs, source hashes, messages and original-source links. The IOS XE casting rows are standards-conflict candidates, not unconditional content-bug findings.

Finding	Pinned source	Rule IDs
O03	MS Windows 10 V3R7 / enhancedV17	SV-220698r991589, SV-220699r991589

Finding	Pinned source	Rule IDs
O03	MS Windows 11 V2R10 / enhancedV18	SV-253255r1210279, SV-253256r1210280
O03	MS Windows Server 2016 V2R8 / enhancedV19	SV-224827r991589, SV-224865r991589
O03	MS Windows Server 2019 V3R9 / enhancedV19	SV-205848r991589, SV-205856r991589
O03	MS Windows Server 2022 V2R10 / enhancedV19	SV-254246r1210256, SV-254283r991589
O03	MS Windows Server 2025 V1R1 / enhancedV1	SV-278031r1182046
O04	MS DotNet Framework 4-0 V2R8 / enhancedV15	SV-225229r1188309
O04	MS Windows 10 V3R7 / enhancedV17	SV-220723r991589
O04	MS Windows 11 V2R10 / enhancedV18	SV-253280r991589
O04	MS Windows Server 2016 V2R8 / enhancedV19	SV-224842r991589
O04	MS Windows Server 2019 V3R9 / enhancedV19	SV-205852r991589
O04	MS Windows Server 2022 V2R10 / enhancedV19	SV-254261r1210262
O04	MS Windows Server 2025 V1R1 / enhancedV1	SV-278008r1211146
O05	MOZ Firefox Windows V6R8 / enhancedV13	SV-251550r961194
O05	MS DotNet Framework 4-0 V2R8 / enhancedV15	SV-225228r1043178, SV-225232r1050651, SV-225233r1117179, SV-225234r1188289, SV-225237r1043178
O05	Oracle Linux 7 V3R3 / enhancedV16	SV-221655r958390
O05	Oracle Linux 8 V2R9 / enhancedV19	SV-248528r958390
O05	Oracle Linux 9 V1R4 / enhancedV5	SV-271689r1091779
O05	RHEL 8 V2R8 / enhancedV22	SV-230226r1069298
O05	RHEL 9 V2R9 / enhancedV13	SV-270174r1044831
O06	MS IIS 8-5 Server V2R7 / enhancedV13	SV-258446r928857
O06	MS IIS 8-5 Site V2R9 / enhancedV13	SV-258445r928856
O06	Oracle Linux 9 V1R4 / enhancedV5	SV-271451r1137691
O06	RHEL 9 V2R9 / enhancedV13	SV-257789r1137691
O07	MS IIS 8-5 Site V2R9 / enhancedV13	SV-214451r879567, SV-214452r879568, SV-214463r879643, SV-214495r879587
O10	Cisco IOS-XE Router NDM V3R5 / enhancedV18	SV-215807r1186318, SV-215812r1137875, SV-215833r961068, SV-215854r1156415
O11	Cisco IOS-XE Router NDM V3R5 / enhancedV18	SV-215833r961068
O12	Cisco IOS-XE Router NDM V3R5 / enhancedV18	SV-215833r961068

Appendix C — original primary blocker accounting

This is the disjoint stage-one primary disposition table, not the overlapping O-number finding inventory. Later validation findings are additive evidence; the frozen DRAFT ledger has not been rewritten.

Primary disposition	Automated Rules
Assessment contracts	1,630
Catalog bindings	992
Effective deprecated original Tests	151
Nonstandard sqlext packages	137
Recorded original validity/deprecation exclusions	32
Further converter/source investigation	39
Total blocked automated Rules	2,981

O08 includes 151 primary effective-Test exclusions plus 11 additional distinct Rules with observed original deprecation diagnostics. O02 shares Rules with O01. O10 shares SV-215833 with O11/O12. These findings must not be summed. The 1,653 manual/OCIL-only Rules are preserved content, not automatic defects.